C. Strong Password

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1400 | Tags: binary search, dp, greedy, strings | https://codeforces.com/problemset/problem/1845/C

Monocarp finally got the courage to register on ForceCoders. He came up with a handle but is still thinking about the password.

He wants his password to be as strong as possible, so he came up with the following criteria:

- the length of the password should be exactly m;
- the password should only consist of digits from 0 to 9;
- the password should not appear in the password database (given as a string s) as a **subsequence** (not necessarily contiguous).

Monocarp also came up with two strings of length m: l and r, both consisting only of digits from 0 to 9. He wants the i-th digit of his password to be between l_i and r_i, inclusive.

Does there exist a password that fits all criteria?

Input

The first line contains a single integer t ($1 \le t \le 10^4$) — the number of testcases.

The first line of each testcase contains a string s ($1 \le |s| \le 3 \cdot 10^5$), consisting only of digits from 0 to 9 — the password database.

The second line contains a single integer m ($1 \le m \le 10$) — the required length of the password.

The third line contains a string l ($|l| = m$), consisting only of digits from 0 to 9 — the lower restriction on each digit.

The fourth line contains a string r ($|r| = m$), consisting only of digits from 0 to 9 — the upper restriction on each digit. $l_i \le r_i$ for all i from 1 to m.

The sum of lengths of s over all testcases doesn't exceed $3 \cdot 10^5$.

Output

For each testcase, print "YES" if there exists a password that fits all criteria. Print "NO" otherwise.

Example

Input

```
5
88005553535123456
2
50
56
123412341234
3
111
444
1234
4
4321
4321
459
2
49
59
00010
2
10
11
```

Output

```
YES
NO
YES
NO
YES
```

Note

In the first testcase, Monocarp can choose password "50". It doesn't appear in *s* as a subsequence.

In the second testcase, all combinations of three digits, each of them being from 1 to 4, fit the criteria on *l* and *r*. However, all of them appear in *s* as subsequences. For example, "314" appears at positions [3, 5, 12] and "222" appears at positions [2, 6, 10].

In the third testcase, Monocarp can choose password "4321". Actually, that is the only password that fits the criteria on *l* and *r*. Luckily, it doesn't appear in *s* as a subsequence.

In the fourth testcase, only "49" and "59" fit the criteria on *l* and *r*. Both of them appear in *s* as subsequences.

In the fifth testcase, Monocarp can choose password "11".